From Advisor to Voting Teammate: Institutional Authority and Information Structures of AI Agents in Bounded-Rational Human Groups

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INTRODUCTION

- **Background:** AI is transitioning from a peripheral assistant to a teammate and engage in human decision-making → How AI is institutionally situated, beyond just whether it is present, may determine whether it amplifies or undermines collective intelligence.
- **Bounded Rationality & Trust:** Due to cognitive limitations and task complexity, human agents suffer from trust miscalibration, a tendency to inappropriately rely on or dismiss AI signals.
- **Social Amplification:** In group settings, individual trust issues are magnified by informational influence and normative influence.
- **False Convergence:** These pressures can lead to a unanimous but incorrect consensus. Consequently, the institutional structure, who accesses the AI and how its influence is moderated, becomes the critical safeguard for collective intelligence.
- **The Research Gap:** We bridge the gap between micro-level cognitive biases and macro-level collective outcomes by exploring how institutional regimes either mitigate or amplify these risks.

RESEARCH QUESTIONS

- How do allocations of AI authority within human groups shape collective accuracy, and what are the trade-offs in cascade risk?
- How do AI voting weights interact with institutional regimes to shape performance, and do critical thresholds emerge?
- Through which pathways do institutional regimes produce their effect?

METHOD

Tool: NetLogo Agent-Based Model (1,113,600 runs)

Dimensions:

- Access structure: centralised or distributed
- Authority type: advisory, voting, or hybrid

Regimes:

no-AI baseline, advisor-centralised, advisor-distributed, pure-voter, hybrid-centralised, and hybrid-distributed

Agent Parameters:

- Humans: signal quality, AI trust, conformity propensity, confidence
- AI: task accuracy, vote weight (w_{ai})

Belief updating:

- private signal + AI recommendation (trust-weighted) + informational influence + normative influence

Measures: collective accuracy, error cascade rate, false convergence

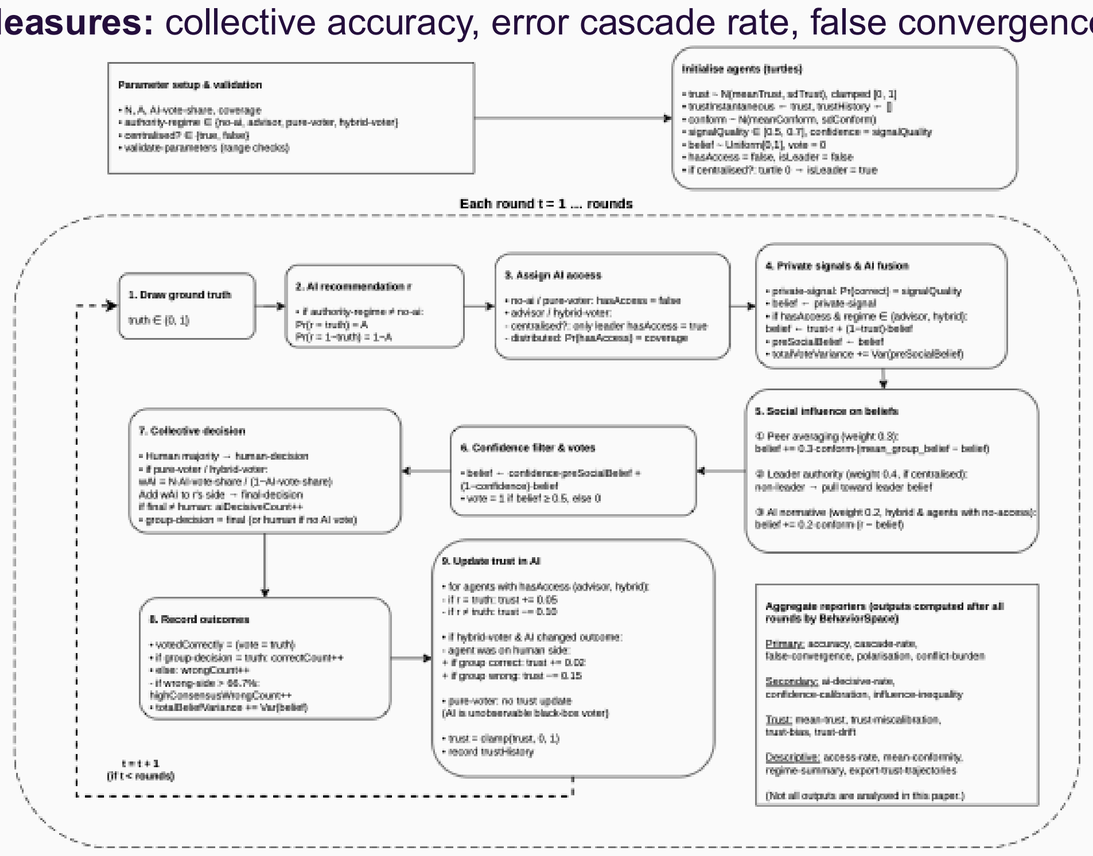

Figure 1: Agent-Based Model procedure.

SIMULATED RESULTS

- **Accuracy ranking:** advisor-centralised (0.824) > no-AI baseline (0.797) > advisor-distributed (0.794) > pure-voter (0.787) > hybrid-centralised (0.786) > hybrid-distributed (0.766)
- **Error cascade:** Centralised & pure-voter: low cascade rates (0.069-0.080) Distributed: six times higher (0.447-0.480);
- Mechanical aggregation introduces a distinct low-consensus error pathway: hybrid-distributed showed lower cascade rate than advisor-distributed yet worse accuracy.
- **Vote weight (w_{ai}):** Beyond $w_{ai} \approx 0.25$-0.30, accuracy drops (0.797→ 0.766), and further to 0.752 at 0.425; false convergence decreases but overall accuracy does not improve.

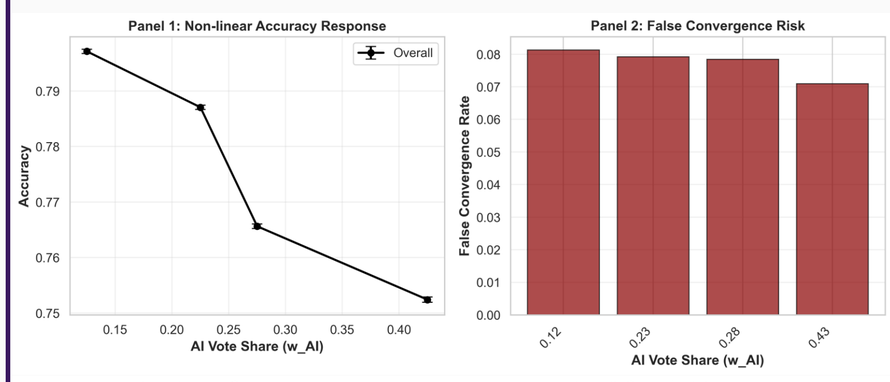

Figure 2: Error cascade mechanism: cascade rates by regime and the relationship between cascade rate and accuracy.

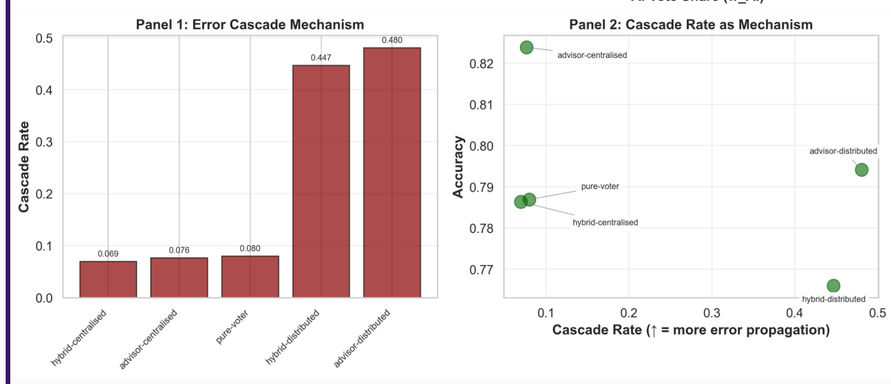

Figure 3: Effect of AI vote weight (w_{ai}) on accuracy and false convergence.

DISCUSSION

Implications: How AI is situated matters.

- The "Circuit Breaker" Effect: In Centralised-Advisory regimes, the leader acts as a human "circuit breaker," filtering AI suggestions through their own judgment before the signal can trigger a group-wide conformity cascade.
- Rethinking "AI Democratisation": Distributed access is beneficial only if human accuracy is high; otherwise, it systematically harms collective intelligence by fueling informational cascade.
- Granting AI voting power introduces normative conformity pressure on top of informational influence, compounding errors, because it might encourage people to suppress their private signals even when they observe disagreement with the AI.

Limitations: Parameters are theoretically assumed.

- Behavioural parameters informed by theories and intuitions rather than empirically calibrated values → accuracy figures are ordinary rankings under bounded conditions and not point estimates
- The study oversimplified human complexities, interactions, and ways of using AI to relatively arbitrary rules → limited generalisability
- The simulated data only provides tentative results and produces sufficient not necessary explanations and only under bounded pre-conditions.

Future Directions: Test with real teams and calibrate empirically.

- Experically test whether centralised AI-informed committees outperform distributed AI voting with real human teams across multiple conditions in real group decision-making settings.
- Use large-scale behavioural measurement to calibrate dynamic parameters.
- Extend simulations to larger and more structurally complex group settings.

REFERENCES

Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. 2024. Information cascades and social learning. Journal of Economic Literature 62, 3 (2024), 1040–1093.
Dan Braha and Marcus AM de Aguiar. 2025. Generalizing Condorcet's Jury Theorem to Social Networks. arXiv preprint arXiv:2510.16808 (2025).
Morton Deutsch and Harold B Gerard. 1955. A study of normative and informational social influences upon individual judgment. The journal of abnormal and social psychology 51, 3 (1955), 629.
Berkeley J Dietvorst, Joseph P Simmons, and Cade Massey. 2015. Algorithm aversion: people erroneously avoid algorithms after seeing them err. Journal of experimental psychology: General 144, 1 (2015), 114.
Suyash Fulay, Sercan Demir, Galen Hines-Pierce, Hélène Landemore, and Michiel Bakker. 2025. Shareholder Democracy with AI Representatives. arXiv preprint arXiv:2510.23475 (2025).
Mo Hu, Guanglu Zhang, Leah Chong, Jonathan Cagan, and Kosa Goucher-Lambert. 2025. How being outvoted by AI teammates impacts human-AI collaboration. International Journal of Human–Computer Interaction 41, 7 (2025), 4049–4066.
Bryan D Jones. 1999. Bounded rationality. Annual review of political science 2, 1 (1999), 297–321.
Ines Lindner. 2008. A generalization of Condorcet's Jury Theorem to weighted voting games with many small voters. Economic Theory 35, 3 (2008), 607–611.
Jennifer M Logg, Julia A Minson, and Don A Moore. 2019. Algorithm appreciation: People prefer algorithmic to human judgment. Organizational Behavior and Human Decision Processes 151 (2019), 90–103.
Kazuo Okamura and Seiji Yamada. 2020. Adaptive trust calibration for human-AI collaboration. Plos one 15, 2 (2020), e0229132.
Kexin Quan, Dina Albassam, Mengke Wu, Zijian Ding, and Jessie Chin. 2025. Towards AI as Colleagues: Multi-Agent System Improves Structured Professional Ideation. arXiv preprint arXiv:2510.23904 (2025).
Uri Wilensky. 1999. NetLogo. Evanston, IL. http://ccl.northwestern.edu/netlogo/
Rui Zhang, Wen Duan, Christopher Flathmann, Nathan McNeese, Guo Freeman, and Alyssa Williams. 2023. Investigating AI teammate communication strategies and their impact in human-AI teams for effective teamwork. Proceedings of the ACM on Human-Computer Interaction 7, CSCW2 (2023), 1–31.